

Lending Club Leakage Audit — Dictionary-Verified + Data-Verified (Task A2 / Contribution 1)

Status: All 151 columns classified, cross-checked against the official Lending Club Data Dictionary (Approved + Notes), and the two genuinely ambiguous fields verified empirically against the loaded data (2,260,701 loans). Dictionary confirmations are marked [**dict**]; empirical confirmations are marked [**data**].

Purpose: classify every column by whether the information was available **at loan origination**. Only origination-time fields may be used as model features. Post-origination fields encode the outcome and cause leakage (the reason many published LC papers report $AUC > 0.9$; clean features give ~ 0.68 – 0.73).

Three kinds of DROP (worth distinguishing in Chapter 3): (1) **leakage** — recorded after origination, encodes the outcome; (2) **identifier** — unique label with no predictive meaning; (3) **free text** — unstructured text not used as a feature.

Note on missingness (separate from leakage): the detailed bureau fields (#64 onward) and the `(sec_app_*)` joint-applicant fields were added to LC's data later and/or only apply to joint applications, so they are heavily missing in pre-2012 vintages. KEEP them in principle, but decide per-vintage whether to include them; missingness is handled after this audit.

SPECIAL — target and time fields (not features, but essential)

#	field	dictionary description	role
16	issue_d	"The month which the loan was funded"	VINTAGE / time index for temporal splits — not a feature
17	loan_status	"Current status of the loan"	OUTCOME → default TARGET — not a feature
20	desc	"Loan description provided by the borrower"	free text; out of scope (the "When Words Sweat" angle) — not used

KEEP — origination-time features (model inputs), dictionary-confirmed

#	field	dictionary description (paraphrased)	note
3	loan_amnt	listed amount applied for	available at application
6	term	36 or 60 months	available at application
7	int_rate	interest rate on the loan	KEEP — see discussion (encodes LC's own risk view)
8	installment	monthly payment "if the loan originates"	origination-time
9	grade	LC assigned loan grade	KEEP — see discussion
10	sub_grade	LC assigned subgrade	KEEP — see discussion
12	emp_length	employment length in years	stated at application
13	home_ownership	ownership status at registration / credit report	application-time
14	annual_inc	self-reported	application-

		income at registration	time
15	verification_status	whether income was verified	application- time
21	purpose	borrower's stated loan category	application- time
23	zip_code	first 3 digits from the application	application- time
24	addr_state	state from the application	application- time
25	dti	debt-to- income ratio	application- time
26	delinq_2yrs	30+ dpd incidences in last 2 yrs	bureau at application
27	earliest_cr_line	month earliest credit line opened	bureau at application
28	fico_range_low	FICO at loan origination (low) [dict]	bureau at application
29	fico_range_high	FICO at loan origination (high) [dict]	bureau at application
30	inq_last_6mths	inquiries in past 6 months	bureau at application
31	mths_since_last_delinq	months since last delinquency	bureau at application

32	mths_since_last_record	months since last public record	bureau at application
33	open_acc	open credit lines	bureau at application
34	pub_rec	derogatory public records	bureau at application
35	revol_bal	revolving balance	bureau at application
36	revol_util	revolving utilisation	bureau at application
37	total_acc	total credit lines	bureau at application
54	collections_12_mths_ex_med	collections in 12m ex-medical	bureau at application
55	mths_since_last_major_derog	months since 90-day+ rating	bureau at application
57	application_type	individual or joint	application-time
58	annual_inc_joint	combined co-borrower income	application-time (joint)
59	dti_joint	joint DTI	application-time (joint)
60	verification_status_joint	joint income verified	application-time (joint)
61	acc_now_delinq	accounts now delinquent (bureau pull)	bureau at application

62	tot_coll_amt	total collection amounts ever owed	bureau at application
63	tot_cur_bal	total current balance	bureau at application
64-115	detailed bureau attributes (open_acc_6m, il_util, all_util, avg_cur_bal, bc_util, mo_sin_, num_, pct_tl_nvr_dlq, tot_hi_cred_lim, total_bal_ex_mort, total_bc_limit, total_il_high_credit_limit, etc.)	credit-bureau attributes	KEEP — heavily missing pre-2012 (verify per vintage)
84	delinq_amnt	past-due amount on accounts "now" delinquent	KEEP — "now" = application-time bureau pull [dict]
104	num_tl_120dpd_2m	accounts currently 120 dpd (updated past 2 months)	KEEP — verified [data] (see below)
105	num_tl_30dpd	accounts currently 30 dpd (updated past 2 months)	KEEP — verified [data] (see below)
110	pub_rec_bankruptcies	public-record bankruptcies	bureau at application
111	tax_liens	number of tax liens	bureau at application
116	revol_bal_joint	co-borrower revolving balance	application-time (joint)
117-	sec_app_*	"at time of	application-

128		application for the secondary applicant" [dict]	time (joint) — confirmed
144	disbursement_method	CASH or DIRECT_PAY	set at origination

RESOLVED — formerly ambiguous, now decided

#	field	how resolved	resolution
84	delinq_amnt	dictionary [dict]	KEEP — "now delinquent" refers to the application-time bureau pull
104	num_tl_120dpd_2m	data verified [data]	KEEP — see empirical check below
105	num_tl_30dpd	data verified [data]	KEEP — see empirical check below
7, 9, 10	int_rate / grade / sub_grade	dictionary [dict]	KEEP WITH DISCUSSION — set by LC at origination from its own risk model, so they partly encode a competing PD prediction. Report results with and without them.

Empirical verification of num_tl_30dpd and num_tl_120dpd_2m [data]

The dictionary wording ("currently ... updated in past 2 months") left open whether these reflect the application-time pull or a later refresh. Checked directly in the data:

- **Distribution:** overwhelmingly zero (num_tl_30dpd: 2,184,561 of ~2.19M
non-missing at 0; num_tl_120dpd_2m: 2,105,738 of 0), telling off slightly to

non-missing at 0; num_tl_120dpd_2m: 2,103,758 at 0), falling off quickly to 1, 2, 3+. This is the expected shape of an application-time "accounts currently past due" bureau field — most applicants have none.

- **Missingness:** 70,309 (~3%) and 153,690 (~7%), concentrated in early vintages (consistent with late-added bureau fields, not leakage).
- **Relationship to outcome (the decisive test):** mean values are near-identical for Fully Paid vs Charged Off loans —
 - num_tl_30dpd: Fully Paid 0.00334 vs Charged Off 0.00374
 - num_tl_120dpd_2m: Fully Paid 0.00080 vs Charged Off 0.00090

Only a *very slight* elevation for defaulters, as expected of a mild application-time risk signal. No dramatic gap that would indicate the field was refreshed to encode the loan's later trouble.

Conclusion: both behave as genuine application-time bureau fields. **Retained.**

Chapter 3 sentence earned: "Two fields with ambiguous dictionary wording (`num_tl_30dpd`, `num_tl_120dpd_2m`) were verified empirically; their distributions and near-identical means across paid and defaulted loans confirmed they reflect the application-time credit pull, and they were retained."

DROP — post-origination (leakage), identifiers, or free-text

#	field	kind	dictionary text → why dropped
1	id	identifier	"unique LC assigned ID" — no predictive meaning; possible time-proxy
2	member_id	identifier	"unique LC assigned Id for the borrower" — also all-NA here

4	funded_amnt	leakage	"amount committed at that point in time" — post-decision
5	funded_amnt_inv	leakage	investor-funded amount — post-decision
11	emp_title	free text	"job title supplied" — note [dict]: replaced Employer Name after 9/23/2013 (inconsistent over time)
18	pymnt_plan	leakage	"if a payment plan has been put in place" — later servicing
19	url	identifier	"URL for the LC listing" — not a feature
22	title	free text	"loan title provided by the borrower" — use purpose instead
38	initial_list_status	other	listing status W/F — operational, not borrower risk; drop (optional)

39	out_prncp	leakage	" Remaining outstanding principal" — updates over loan life
40	out_prncp_inv	leakage	remaining outstanding principal (investors) — same
41	total_pymnt	leakage	"Payments received to date " — encodes outcome
42	total_pymnt_inv	leakage	payments received to date (investors) — encodes outcome
43	total_rec_prncp	leakage	"Principal received to date " — encodes outcome
44	total_rec_int	leakage	"Interest received to date " — encodes outcome
45	total_rec_late_fee	leakage	"Late fees received to date " — encodes outcome

			outcome
46	recoveries	leakage	" post charge off gross recovery" — DEFINES default
47	collection_recovery_fee	leakage	" post charge off collection fee" — DEFINES default
48	last_pymnt_d	leakage	"Last month payment was received" — post-origination
49	last_pymnt_amnt	leakage	"Last total payment amount received" — post-origination
50	next_pymnt_d	leakage	"Next scheduled payment date" — post-origination
51	last_credit_pull_d	leakage	"most recent month LC pulled credit" — post-origination
52	last_fico_range_high	leakage	"borrowers last FICO pulled" — post-origination

			origination
53	last_fico_range_low	leakage	"borrowers last FICO pulled" — post-origination
56	policy_code	other	admin/constant (publicly-available flag)
83	chargeoff_within_12_mths	leakage	"charge-offs within 12 months" — outcome-adjacent; drop to be safe
129-143	hardship_* (flag, type, reason, status, deferral_term, amount, start/end dates, length, dpd, loan_status, payment_plan_start_date, orig_projected_additional_accrued_interest, hardship_payoff_balance_amount, hardship_last_payment_amount)	leakage	hardship-plan data — only exists if the loan struggled after origination
145	debt_settlement_flag	leakage	"borrower, who has charged-off , working with settlement company" — post-default
146	debt_settlement_flag_date	leakage	most recent date flag was set — post-origination
147-151	settlement_* (status, date, amount, percentage, term)	leakage	settlement details — only exist post-default

The most dangerous (produce the fake ~0.95 AUC): `recoveries`, `collection_recovery_fee`, `total_rec_prncp`, `total_pymnt`, `last_pymnt_amnt`, `last_fico_range_*`, and the `hardship_*` / `settlement_*` blocks. Each is a direct fingerprint of the outcome.

Summary after dictionary + data verification

- **No major corrections required** — the original audit was sound. The dictionary confirmed every dangerous drop; the two genuinely ambiguous fields were resolved by direct data checks.
- **KEEP (features):** application + bureau fields above (~85 columns; fewer in practice once heavily-missing late-added bureau fields are excluded for early vintages).
- **DROP:** ~50 columns — leakage (payment/recovery/hardship/settlement/last-* fields), identifiers (id, member_id, url), and free text (emp_title, title).
- **SPECIAL:** `issue_d` (vintage), `loan_status` (target), `desc` (out of scope).
- **VERIFIED in data:** `num_tl_30dpd`, `num_tl_120dpd_2m` — retained (behave as application-time bureau fields).
- **DISCUSS in Chapter 3:** `int_rate`, `grade`, `sub_grade` (run models with and without them).

Chapter 3 sentence you've earned: *"Feature timing was classified by hand, verified against the official Lending Club Data Dictionary, and — for two fields with ambiguous wording — confirmed empirically against the data; only fields available at loan origination were retained, and outcome-encoding fields (payments, recoveries, hardship and settlement records, and most-recent credit pulls) were excluded to prevent target leakage."*

Next step: build the `ORIGINATION_FEATURES` vector in R from the KEEP list, then define the default target from `loan_status` and the vintage from `issue_d`.